

OPENENV RL ENVIRONMENT • V0.1

reconcile_gst2b_env

A reinforcement-learning environment for Indian GST input-tax-credit reconciliation, the monthly tax-document-matching workflow that 14 million Indian businesses do by hand.

MEAN COMPOSITE REWARD 0.305 n=5, A100-80GB, Day 1 on-site	LIFT OVER PROMPTED BASELINE +69% +0.125 absolute, vs Qwen2.5-3B	DOCUMENTED FAILURE MODES 5 3 pre-onsite, 2 new on-site findings
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14 million businesses, every month, by hand.

Indian GST law requires every business to reconcile its purchase register against the government-published GSTR-2B file before claiming input-tax-credit. The work is rule-bound, error-sensitive, and almost entirely manual.

PURCHASE REGISTER · BOOKS			FY26 · APR			GSTR-2B · GOVT FILE			FY26 · APR		
INV/0418	ACME PARTS	₹ 18,000	MIS- MATCH ⇔			INV/0418	ACME PARTS	₹ 18,000			
INV/0419	VENDOR-K	₹ 9,200				INV/0419	VENDOR-K	₹ 9,200			
INV/0420	SARASWATI CO	₹ 24,500				INV/0420	SARASWATHI & CO	₹ 24,500			
INV/0421	PRIME LOGI	₹ 6,400				INV/0421	PRIME LOGISTICS	₹ 6,400			
INV/0422	METRO IT	₹ 12,750				— missing —	METRO IT	—			

Five labels · matched, mismatched, only_in_books, only_in_2b, partial_match

Section 36(4) · partial-observability problem

Illustrative example only. Real GSTR-2B contains 30+ columns including HSN/SAC, IRN, place of supply, etc.

Environment and reward design

ENVIRONMENT

action_space	16 typed verbs · 7 query / 7 mutate / 2 meta
labels	5 mark labels per invoice row
observability	partial: ground truth hidden, unit-tested invariant
budget	50 steps per episode, hard cap
structural	-1.0 for submit-before-query
judge	none. no LLM in reward path.

DESIGN COMMITMENT

Every reward signal is arithmetic and reproducible. No model-graded scoring anywhere in the pipeline.

COMPOSITE REWARD · 4 COMPONENTS

ID	SIGNAL	WEIGHT CLAMP
R1	reconciliation_f1	0.40[.01, .99]
R2	itc_delta_accuracy	0.25[.01, .99]
R3	rule_36_4_compliance	0.25[.01, .99]
R4	step_efficiency	0.10[.01, .99]

$R = 0.40 \cdot R1 + 0.25 \cdot R2 + 0.25 \cdot R3 + 0.10 \cdot R4$

Six red-team attacks, all under the ceiling.

RED-TEAM CI · ENFORCED < 0.45

6 attacks · from rewards.py CI · all under 0.45 ceiling

ATTACK	SCORE	CAUGHT BY
query_only	0.349	R1 no labels, R2 no claim
submit_all_matched	0.308	R3 overclaim, R4 budget
overflag_rings	0.283	R1 no mark labels, R2 no claim
submit_all_mismatched	0.266	R1 4-of-5 class F1, R2 zero claim
zero_itc	0.266	R2 claim-vs-true, R3 blind by design
confirm_spam	0.010	R3 no query, R4 budget

max attack score 0.349 · ceiling 0.45 · CI fails the build if any attack ≥ ceiling

POSITION VS PRIOR WORK

VERIFY · ratings vs Gaia2 / AppWorld / τ-bench

PROPERTY	APP GAIA2	T- WORLD	τ- BENCH	THIS
Arithmetic-only reward				
Red-team CI ceiling				
Regulator-published correctness				
Hidden GT, first-class				

● present ◐ partial ○ absent

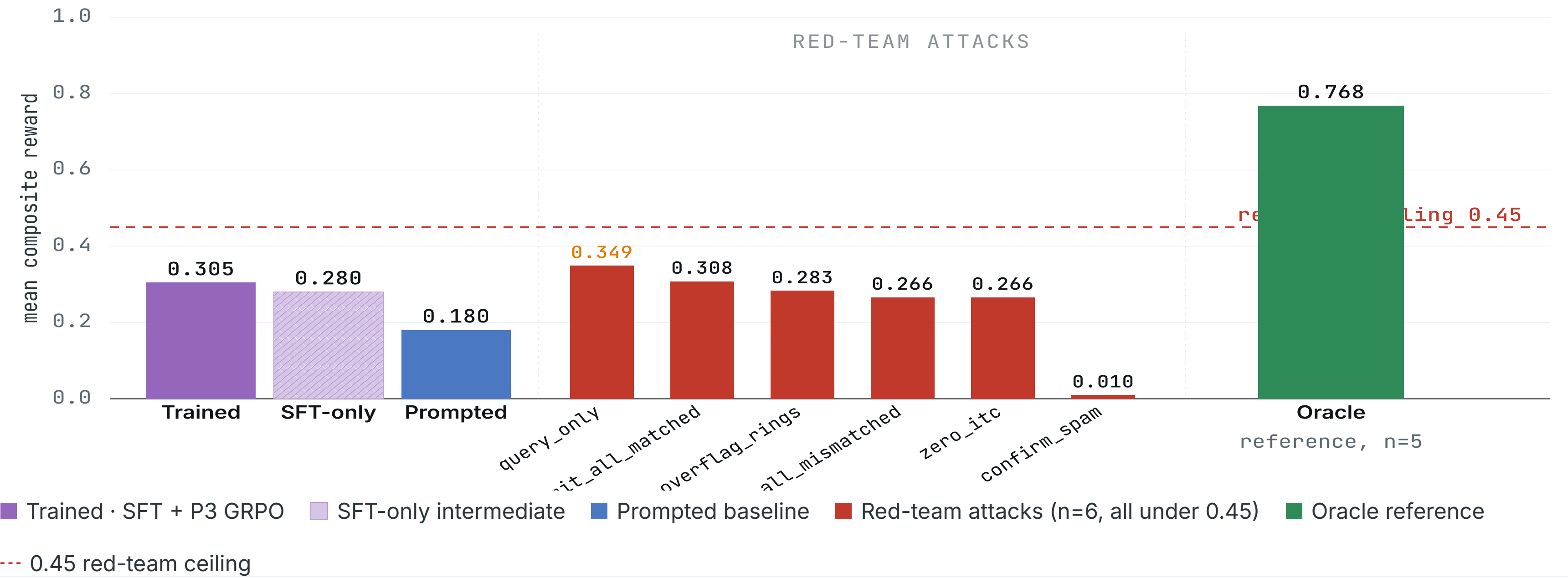
Three failure modes, three model scales.

Before the on-site day, every Qwen3 scale failed in a different way. Each failure pinned a specific reward component and told us what to fix next.

MODEL	OBSERVED BEHAVIOR	REWARD COMPONENT PINNED
Qwen3-0.6B	Bimodal tool-call rate at 0.25, half the rollouts emitted no tool calls at all. tool-call rate 0.25 · entropy 0.44	R1 + R2 floor
Qwen3-1.7B	Entropy collapsed to 0.12, policy degenerated to zero tool calls. tool-call rate 0.00 · entropy 0.12 (collapsed)	R1 + R2 floor
Qwen3-4B	Over-queried the environment, exhausted the 50-step budget before submitting. tool-call rate ~1.00 · R4 floor 0.01	R4 floor

The 4B over-query failure is what made the on-site SFT + GRPO run worth doing: the model could query, it just could not stop.

Day 1 on-site results.



+0.125 absolute over prompted (+69%) · +0.025 GRPO over SFT

n=5, A100-80GB, Day 1 on-site

FAILURE MODE 4

We thought our model was broken. We were wrong.

Four sequential bugs in the audit infrastructure masked real model behavior. Each one had to be fixed before the next became visible.

BUG 1

Greedy decoding

Eval ran at temperature 0, hiding the policy distribution we actually trained.

BUG 2

Fresh context per turn

Eval callback reset history every step. Reported a flat **0.354** across all 100 GRPO steps.

BUG 3

Prompt-format false alarm

Mismatch between train and eval chat templates that turned out not to matter, but cost an afternoon to rule out.

BUG 4 • LOAD-BEARING

Missing tools= param

apply_chat_template was called without the tools list, so the model saw no schema. Once fixed, the real curve appeared.

LESSON

Diagnostic infrastructure is as load-bearing as the model itself. Empirically confirmed across both SFT and GRPO stages.

FAILURE MODE 5

The more-trained checkpoint scored lower.

The same reward design that defends against six red-team attacks also makes a cheap 3-step query_only shortcut score higher than 50 steps of honest marking. Documenting this asymmetry is engineering rigor, not failure.

STEP 350

0.314

training mean

STEP 375 ↓

0.280

training mean

PARTIAL MITIGATION

0.305P3 Tier 2c length-shaping bonus,
+0.025

ARTIFACTS

HF Space `huggingface.co/spaces/akashkathole/reconcile_gst2b_env`

GitHub `github.com/akashkathole7/OpenEnv`
branch: `scaffold/reconcile-gst2b`

Writeup `BLOG.md` • `LESSONS_LEARNED.md`

Demo 90-second YouTube walkthrough

Built honestly.
Tested adversarially.
Shipped open source.

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OPENENV HACKATHON 2026